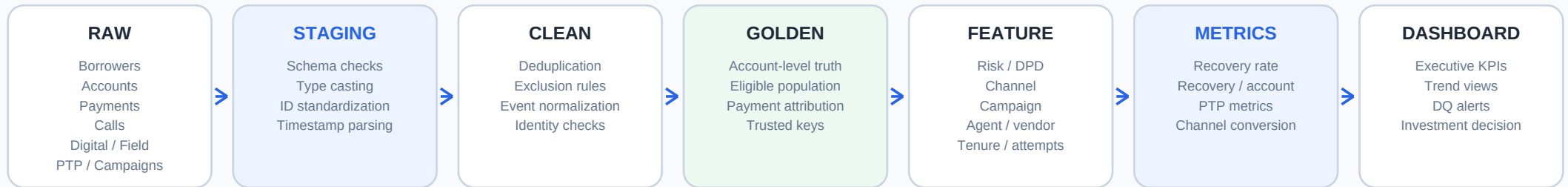


Collections Analytics Architecture

Governed pipeline from source data to executive decisioning



Controls applied across the pipeline

Data contracts

Required fields, schemas, type rules

Freshness

Watermarks, late-arriving event handling

Keys & identity

account_id / payment_id / ptp_id; agent resolution

Backfills

Versioned corrections and reproducible reprocessing

Attribution

Post-PTP timing and promised-date qualification

Monitoring

Volume, nulls, duplicates, timestamp anomalies

Analytical outputs and current controls

Golden layer

- 30,000 accounts
- 0 duplicate account IDs
- 0 missing values

Forensic controls

- 500 duplicate payment rows checked
- PTP attribution window enforced
- Agent/vendor mappings flagged

Leadership layer

- Monthly recovery trend
- Segment & channel views
- ■10 Cr investment guardrails

Design principle: metrics are calculated only after governed cleaning and attribution; monitoring and DQ checks run throughout.